

OUSSAMA YAZBEK-JAMAL

Software Engineer — Full-Stack Development & Applied AI/NLP

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Computer Science graduate with hands-on experience building full-stack systems and integrating applied machine learning — from fine-tuning and deploying transformer/CNN models in a production-shaped platform, to building a rule-based Arabic NLP pipeline from first principles. Comfortable owning a project end-to-end: architecture, database design, API design, and debugging real-world failure modes rather than only ideal-case behavior.

EDUCATION

Bachelor of Science in Computer Science and Information Technology

Lebanese International University (LIU), Bekaa, Lebanon — *Coursework complete, 2026 (certification pending)*

PROJECTS

InfraWatch — Civic Infrastructure Reporting Platform (Senior Project)

- Designed and built a full-stack platform (React, Node.js/Express, PostgreSQL via Supabase) serving four distinct roles — Citizen, Org Staff, Org Lead, Admin — with jurisdiction-scoped data ownership, validated end-to-end across three demonstration regions.
- Integrated three ML components into a live product: XLM-RoBERTa for multilingual report-urgency classification, a fine-tuned MobileNetV2 for road-damage image classification (~0.78 macro F1 on RDD2022), and a hybrid gazetteer + multilingual NER pipeline for Arabic/English/French location extraction from free text.
- Diagnosed and redesigned a fuzzy jurisdiction-matching system after identifying multiple distinct real-world failure modes, unifying the fix into a single shared resolution function reused across citizen and organization workflows.
- Built a conservative, three-signal duplicate-report detection system and a fail-soft AI architecture in which every automated suggestion remains fully overridable by human judgment.
- Implemented a secure, role- and jurisdiction-scoped REST API (60+ endpoints, JWT auth) with authorization middleware distinct from simple role checks — including a claim-on-edit ownership model for shared civic data.

Arabic Morphological Analyzer & Parser — Computational Linguistics Project

- Built an Arabic NLP pipeline from first principles: normalization, tokenization, morphological analysis, lemmatization, root extraction, POS tagging, and full syntactic parsing — no pretrained model in the core pipeline.
- Implemented rule-based extraction of Arabic's non-concatenative root-and-pattern morphology via consonant/vowel structural template matching, covering 10+ verb forms and participle patterns.
- Handled Arabic-specific complexity largely absent from English NLP tooling: broken (irregular) plural lemmatization, pronoun-clitic stripping, and prefix/suffix disambiguation.
- Designed a hybrid rule + ML POS tagger (scikit-learn Logistic Regression) that falls back to a trained classifier only when the rule-based tagger is genuinely uncertain, with a human-readable explanation generated for every tagging decision.
- Implemented dual word-order syntactic parsing (VSO and SVO — both valid in Arabic), producing NLTK-based constituency trees (NP/VP/PP) correctly structured for either order.

TECHNICAL SKILLS

Languages: JavaScript, Python, SQL, HTML/CSS

Frontend: React, React Router, React-Leaflet

Backend: Node.js, Express, REST API design, JWT authentication

Database: PostgreSQL, Supabase

ML / NLP: Hugging Face Transformers (XLM-RoBERTa), CNN fine-tuning (MobileNetV2), scikit-learn, NLTK, PyArabic, Arabic computational morphology

Tools: Git, Postman, Google Maps API, OpenStreetMap/Overpass API